Ruiyuan Bai

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Education

City University of Hong Kong 2025.09–Present
Master of Fine Arts, Human-Computer Interaction Stream Hong Kong
cGPA: 3.66; Expected graduation: May 2027

The Hong Kong Polytechnic University 2021.09–2025.05
BSc (Hons) Financial Technology & Artificial Intelligence – Computing Hong Kong
cGPA: 3.15; Rank: Top 24%
Selected coursework: Programming Fundamentals, Design and Analysis of Algorithms, Software Engineering, Human–Computer Interaction, Artificial Intelligence, Machine Learning.

Montrose School 2018.09–2021.05
High School Medfield, Massachusetts

Publications

*Equal contribution; †Corresponding author.

Sirui Ding*, **Ruiyuan Bai***, Wenqian Wu, Zheng Wang, and Yue Gu†. (2026). Beyond the Gaze: Exploring Visual Attention and Interpretation in Open-Ended Art Viewing. *ACM Multimedia Asia 2026*.

Rui Yao, **Ruiyuan Bai**, Yang Chen, Qiuyuan Ren, Felicia Fang-Yi Tan, Xiaoyu Zhang, and Shengdong Zhao. (2026). Demonstrating PersonaMail: Learning and Adapting Personal Communication Preferences for Context-Aware Email Writing. *IUI '26 Companion*, 160–166. [DOI]

Yue Gu, Yuxuan Li, **Ruiyuan Bai**, Zhiwei Yao, and Shengdong Zhao. (2026). ColoGene: Empowering Designers with Genetic Narratives of Traditional Chinese Colors. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. [DOI]

Ben Stubbs, Nanditha Krishna, **Ruiyuan Bai**, Chensy (Xiaochen) Cao, Sabrina Jiang, Xingru (Winnie) Yi, and Yihan Wang. (2025). It's all fun and games: Contemporary newsgames and journalism. *Australian Journalism Review*, 47(2), 173–196. [DOI]

Selected Research Experience

GlanceMap: Audio Learning on the Move Hong Kong
Research Collaborator, User Studies and Data Analysis 2025.07–2026.09
Synteraction Lab, City University of Hong Kong

- Prepared and refined materials across **14 pilot rounds** for an AR audio-learning interface; analyzed data from five pilots.
- Conducted **two post-experiment interviews per participant**, analyzed formal-study data, and contributed to manuscript writing.
- **Second-author manuscript; under review at CHI 2027.** Rui Yao, **Ruiyuan Bai**, Runze Cai, Luoyin Lin, Nuwan Janaka, and Shengdong Zhao. (2026). GlanceMap: Glanceable Relational Structure for Learning Expository Audio Content on the Move.

Icon Color Design: Clarity, Comfort, and Semantics Hong Kong
Research Collaborator, Experimental Studies 2025.05–2025.09
Synteraction Lab, City University of Hong Kong; Cultural Heritage Lab, Wuhan University

- Conducted experimental sessions across **all three studies** of icon color in optical see-through head-mounted displays; designed color treatments examining visual clarity, comfort, and semantic associations.
- **Second author; manuscript writing; submission to IJHCI in progress.** Yue Gu, **Ruiyuan Bai**, Chao Huang, Runze Cai, Ashwin Ram, and Shengdong Zhao†. (2026). Icon Color Design in Optical See-through Head-Mounted Displays: Balancing Clarity, Comfort, and Semantics. *SSRN preprint*. [Preprint] [DOI]

Ambulance Rush!: A Newsgame on Ambulance Ramping Remote
IVE Virtual Intern; Team Leader and Lead Programmer 2024.03–2025.03
Australian Research Centre for Interactive and Virtual Environments (IVE)
University of South Australia (now Adelaide University)

- Recruited and led the development team, coordinated gameplay design, and programmed game systems in Unity/C#.
- Interviewed students and ran playtests to inform iterative development; introduced ambulance ramping in Adelaide to **over 2,000 players**. [Play game]
- Co-authored a study of storytelling in newsgames with Dr. Ben Stubbs; **third author**, *Australian Journalism Review* (2025), listed above.

Game Development

Her Sound Resound Beijing, China
Narrative Writer and Programmer 2026.08

- **Second Place, Audience Choice Award**, Beijing 1,000-Person Women's Hackathon. [Play game]

Earth Is Not Humanity's Alone Remote
Designer and Developer 2026.07

- **Selected for exhibition**, 16th China International Digital Publishing Expo. [View project]

Forest Trending Shenzhen, China
Independent Developer 2026.06

- **Outstanding Project**, Tsinghua University's Agentland Fortnight.

Taste of Connection Hong Kong
Unity/C# Programmer and Production Coordinator 2025.12–2026.04

- Developed a Unity 2D game exploring work–family choices through Chinese dining culture, with two story routes and six levels.

Who Survives to the End Beijing, China
Co-team Leader and Programmer 2023.07–2024.02

- **Gold Award**, 8th China Association for Educational Technology Competition.

Internship Experience

Sina Corporation 2024.05–2024.07
Intern, Automotive Marketing Technology Department; Beijing, China

Evaluated and compared LLM APIs for automotive marketing applications; developed task-specific prompts to explore their use in marketing workflows.

Skills

Research methods: User-study execution, experimental material development, pilot testing, semi-structured interviews, and playtesting.

Data analysis & research writing: Quantitative and qualitative analysis of user-study data; academic manuscript preparation.

Programming & game development: Python, C#, Lua; Unity and Roblox.

Web & interface prototyping: JavaScript/TypeScript, HTML/CSS, React, Next.js, Node.js/Express; Figma.

LLM applications: Model API evaluation and task-specific prompt design.